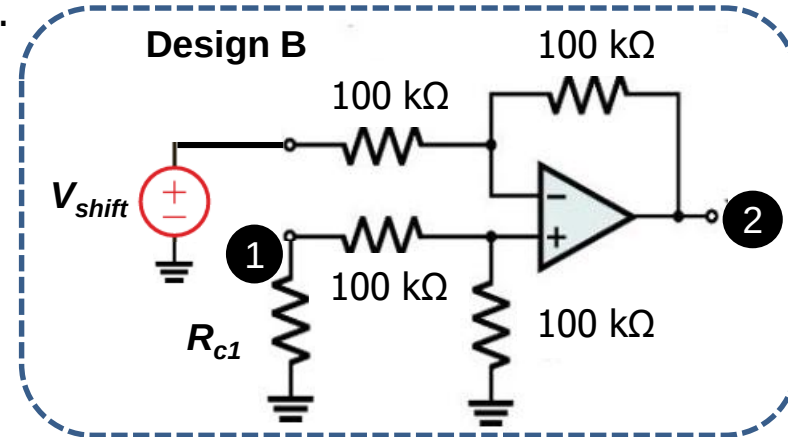
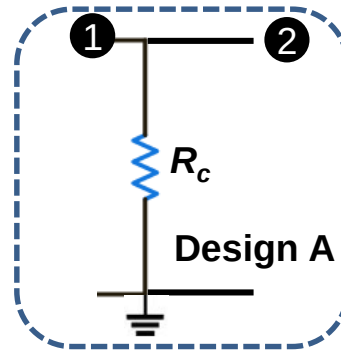
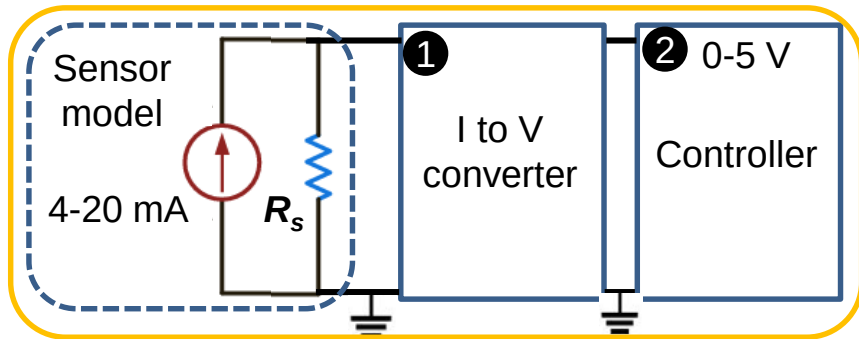


Tutorial Quiz-7 (Max. Duration: 20 mins)

Instructions: Please show your work clearly and specify your assumptions (if any).

A factory sensor outputs a current signal ranging from 4 mA to 20 mA ($R_s = 500 \Omega$). However, your microcontroller can only read voltages between 0 V and 5 V. You must design a circuit to convert this current into the correct voltage (i.e. 4 mA $\rightarrow$ 0 V and 20 mA $\rightarrow$ 5V). You are given two options: Design A and Design B.



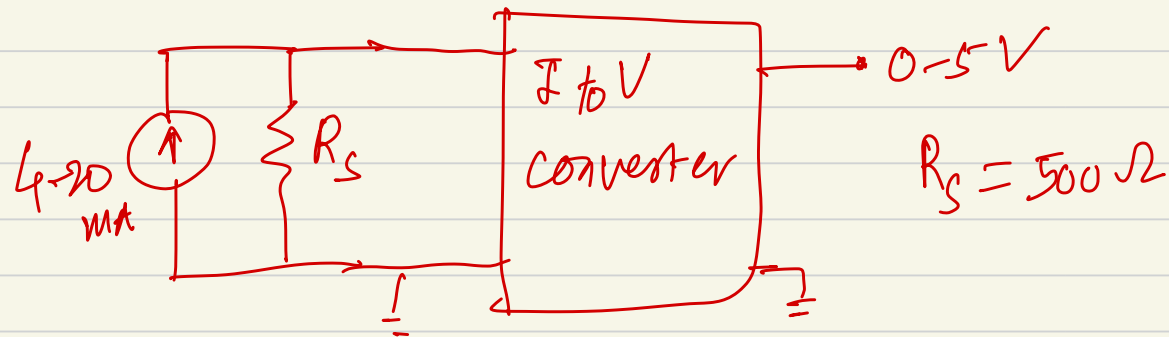
For Design A:

1. (2 points) Calculate the resistor value (R_c) needed to output exactly 5 V when the sensor current is 20 mA.
2. (2 points) Using the resistor you just calculated, what voltage will Design A output when the current drops to 4 mA? Briefly explain why this makes Design A not an appropriate choice.

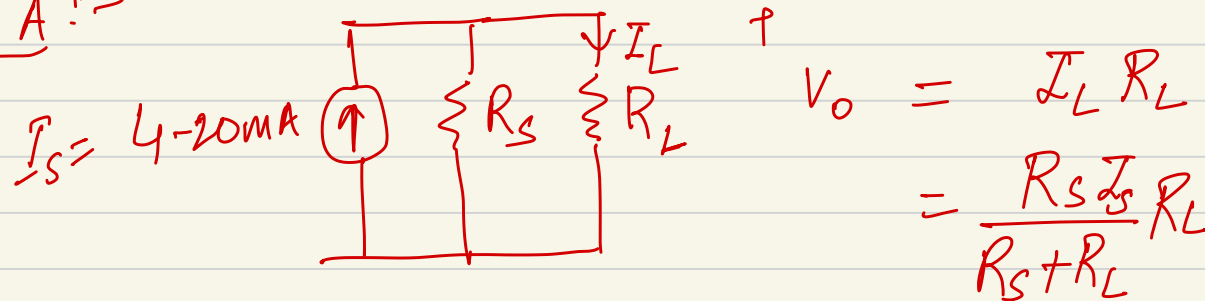
For Design B:

3. (3 points) Calculate the exact voltage shift needed to perfectly map 4 mA to 0 V, and 20 mA to 5 V [Hint: taking the effective resistance at node 1].
4. (3 points) Calculate the required value of R_{c1} if R_1 is chosen to be 100 k Ω . Comment on the choice of R_1 .

Tutorial Quiz-7



Design A :-



$$V_O = I_L R_L$$
$$= \frac{R_S I_S R_L}{R_S + R_L}$$

Q1. $V_O = 5V$ when $I_S = 20mA$

$$\Rightarrow \frac{5V}{20mA} = \frac{R_S R_L}{R_S + R_L} \Rightarrow \frac{R_S R_L}{R_S + R_L} = \frac{5 \times 1000}{20} = 250$$

if $R_S = 500 \Omega$, $R_L = \underline{500 \Omega}$

Q2. If $I_S = 4 \text{ mA}$, $V_O = \frac{R_S R_L}{R_S + R_L} \times 4 \text{ mA} = \frac{500 \times 500}{1000} \times 4 \text{ mA} = 1 \text{ V}$

→ We require 0V corresponding to 4mA, however we get 1V.

Design B

For difference Amplifier:-

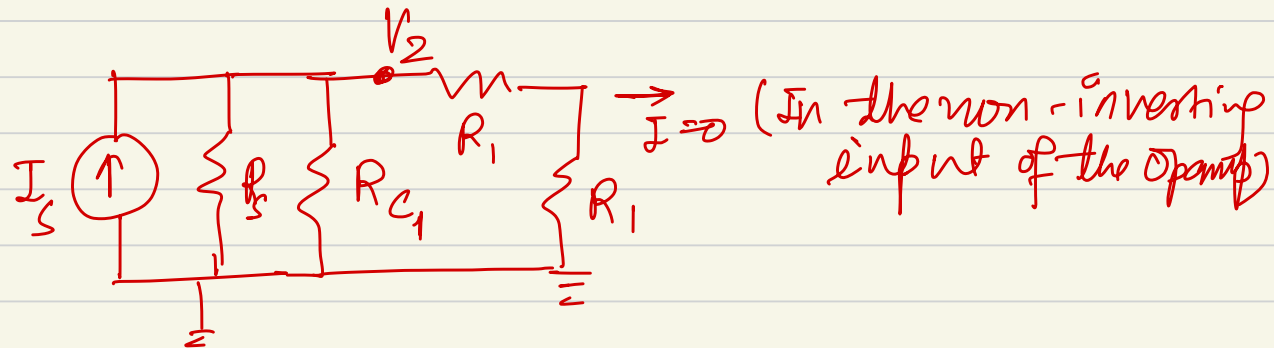
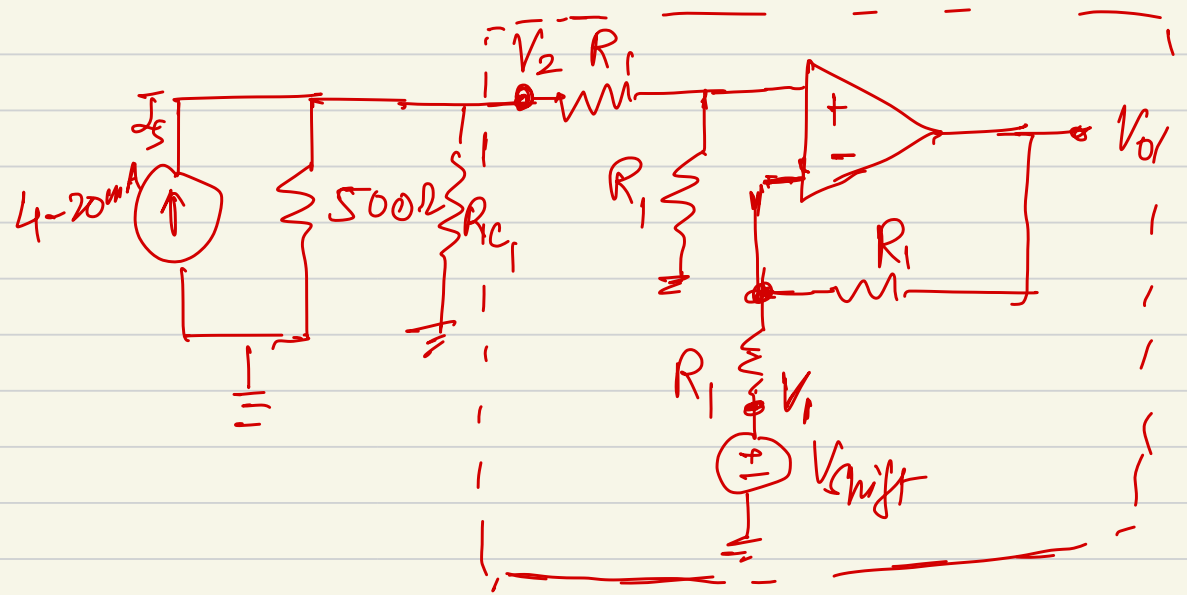
$$V_O = \frac{R_1}{R_1} (V_2 - V_1)$$

here $V_1 = V_{\text{shift}}$

For V_2

$$V_2 = I_S \cdot R_{\text{eff}}$$

$$R_{\text{eff}} = R_S \parallel R_{C1} \parallel 2R_1$$



$$\Rightarrow V_o = I_s R_{eff} - V_{shift}$$

$\Rightarrow$ for V_o to be zero at 4 mA,

$$0 = R_{eff} \times 4 \text{ mA} - V_{shift} \quad \text{--- (1)}$$

for V_o to be 5 V at 20 mA

$$\Rightarrow 5 \text{ V} = R_{eff} \times 20 \text{ mA} - V_{shift} \quad \text{--- (2)}$$

Subtracting (2) - (1) $\Rightarrow$

$$\Rightarrow \frac{5 - 0}{(20 - 4) \text{ mA}} = R_{eff} \Rightarrow R_{eff} = \frac{5}{16} \text{ k}\Omega = 312.5 \Omega$$

$$\text{Q3. } V_{shift} = R_{eff} \times 4 \text{ mA} = \frac{5}{16 \text{ mA}} \times 4 \text{ mA} = \frac{5}{4} = 1.25 \text{ V}$$

Q4.

$$R_{eff} = R_s \parallel R_{C1} \parallel 2R_1, \quad R_s = 500 \Omega, \quad 2R_1 = 200 \text{ k}\Omega$$

$$\Rightarrow \frac{16}{5000} = \frac{1}{500} + \frac{1}{R_{C1}} + \frac{1}{200 \text{ k}} \Rightarrow \frac{1}{R_{C1}} = \frac{16}{5000} - \frac{1}{500} - \frac{1}{200 \text{ k}}$$

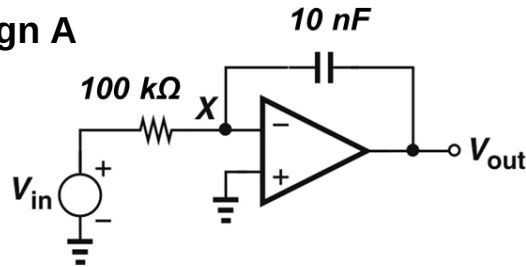
$$\Rightarrow \boxed{R_{C1} = 836.82 \Omega}$$

Tutorial Quiz-8 (Max. Duration: 20 mins)

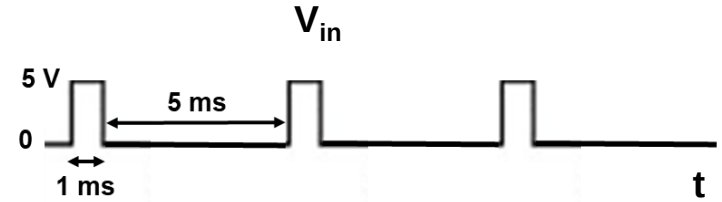
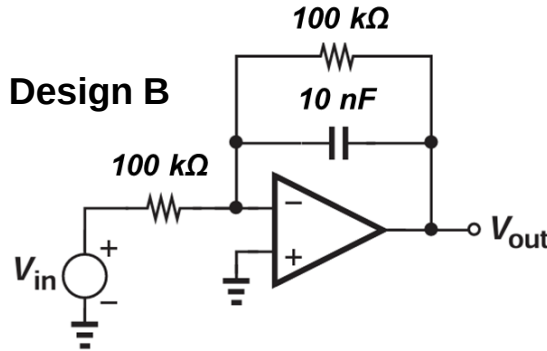
Instructions: Please show your work clearly and specify your assumptions (if any).

You are designing an artificial neural network using analog circuits. A biological neuron receives short electrical voltage spikes from its neighbors. It adds (integrates) these spikes together. If the internal voltage reaches a specific threshold, the neuron fires. You are testing two different Op-Amp circuits (Design A and Design B) to act as the body of the neuron.

Design A



Design B

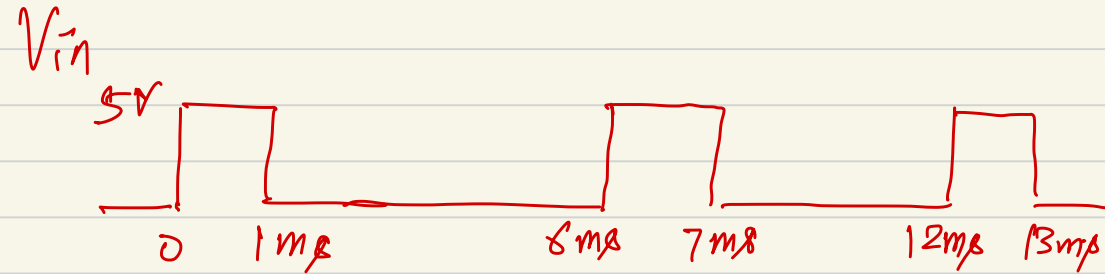
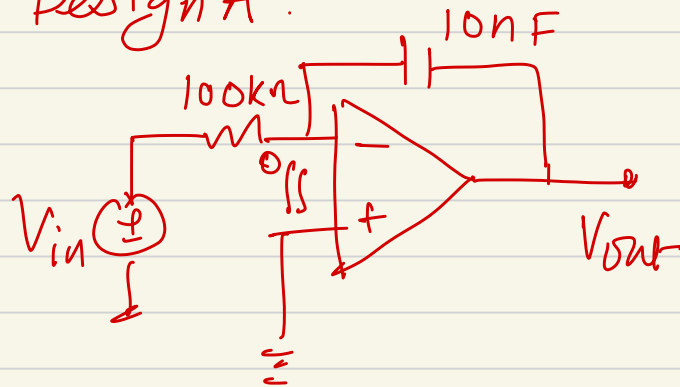


Questions:

1. For Design A, (3 points) calculate the exact output voltage at the end of the very first 1 ms spike.
(3 points) Sketch the output waveform (V_{out} vs. Time) for the entire sequence of all three spikes [Indicate voltage levels].
2. (3 points) For Design B, sketch the *approximate* output waveform for all three input spikes [Need not indicate the voltage levels].
3. (1 point) Real biological neurons exhibit temporal summation. It means they only fire if multiple spikes arrive rapidly one after the other. If the spikes arrive too slowly, the neuron forgets them and will not fire. Which design (A or B) correctly models this biological behavior?

Tutorial Quiz-8

Q. 1. Design A:



at $t = 0$, $V_{out} = 0V$ (capacitor is initially discharged)

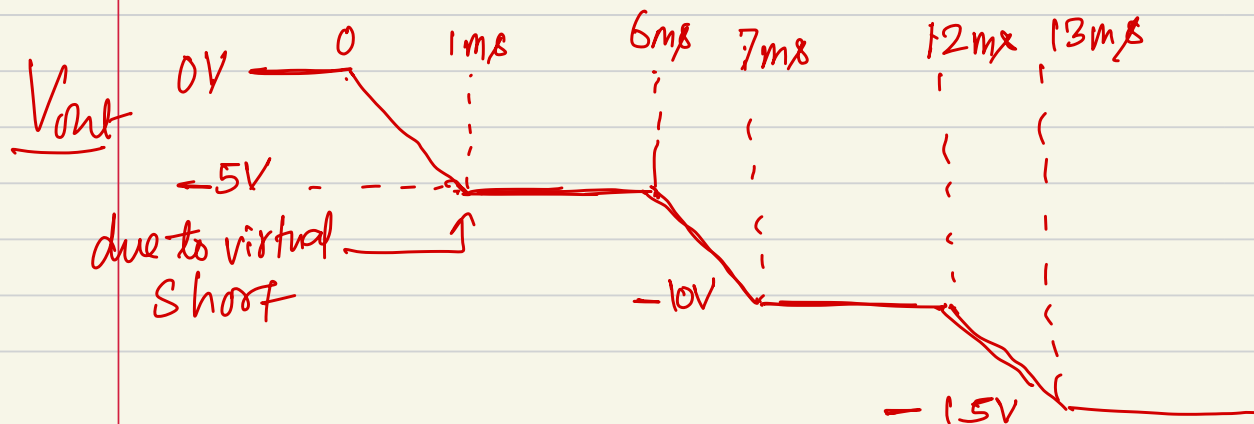
$$V_{out} = -\frac{1}{RC} \int V_{in}(t) dt \Rightarrow$$

$$= -\frac{1}{RC} \cdot V_{max} t + V_{out}(0)$$

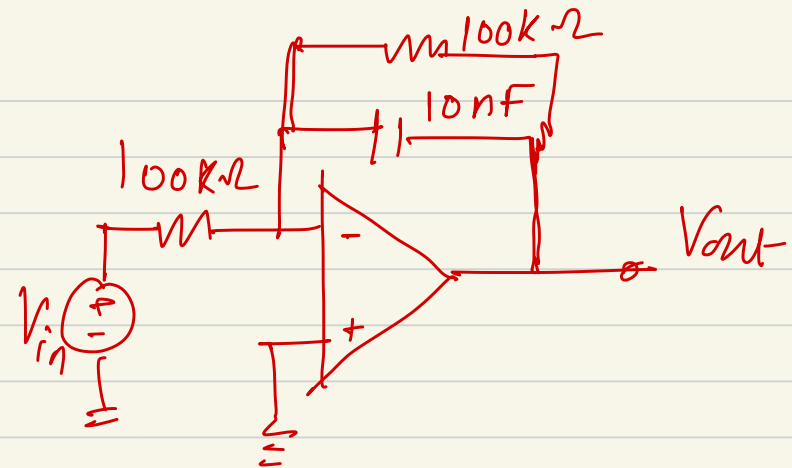
$$V_{out}(t = 1ms) = -\frac{1}{(100k \times 10 \times 10^{-9})} \int_0^{1ms} 5V dt$$

$$= -\frac{1}{1ms} \cdot 5V (1ms)$$

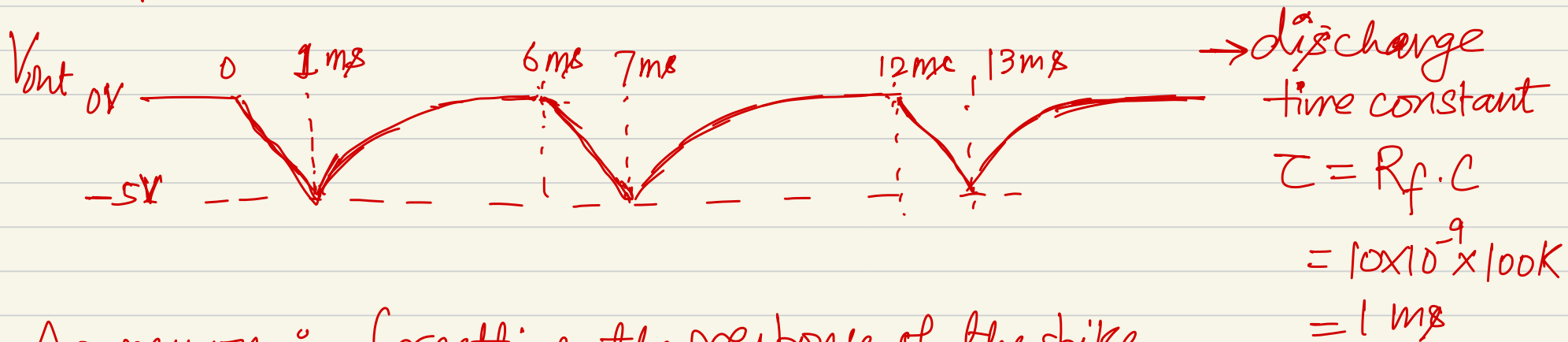
$$= -5V$$



Q 2. Design B



as there is a feedback resistor present, when $V_{in} = 0V$, if capacitor is charged, it can now discharge through feedback resistor therefore, the output will not remain constant when $V_{in} = 0V$.



Q 3. As neuron is forgetting the response of the spike, it is similar to the discharging of the capacitor through feedback resistor. So, it is design B.

Tutorial Quiz-9 (Max. Duration: 20 mins)

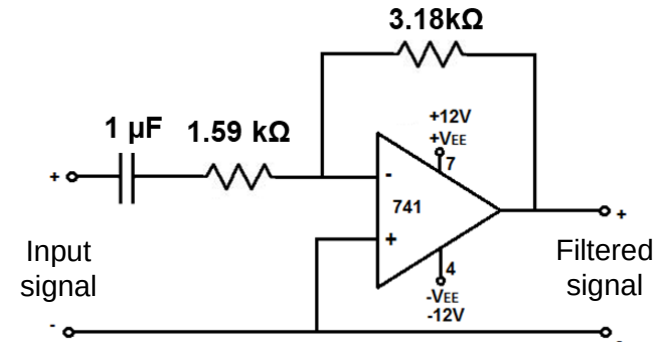
Instructions: Please show your work clearly and specify your assumptions (if any).

You are building a diagnostic tool to check if a factory motor is breaking down. A vibration sensor attached to the motor outputs a mixed voltage signal. This input signal contains two parts: $V_{in}(t) = 1.0\sin(2\pi \cdot 2 \cdot t) + 1.5\sin(2\pi \cdot 2000 \cdot t)$, where 2 Hz is an unwanted signal while 2 kHz is what we want to measure.

Questions:

1. To get rid of the unwanted rumble, the raw signal enters is passed through the circuit shown in the figure.

(5 points) Calculate the cutoff frequency (f_c) of this filter and the maximum positive peak voltage of the filtered output waveform.



2. The filtered signal is passed through a Sample & Hold peak-detector. This circuit **grabs only the maximum positive peak voltage** you calculated in Part 1 and holds it flat as a steady DC voltage.

This steady DC voltage is fed into a 2-bit ADC. The ADC has an acceptable voltage range of 0 V to 4 V.

A. (1 point) Calculate the resolution (step size) of this 2-bit ADC.

B. (1 point) Based on the held DC voltage, determine the exact 2-bit digital output generated by the ADC. (*Assume standard binary encoding where 0 V = 00*)

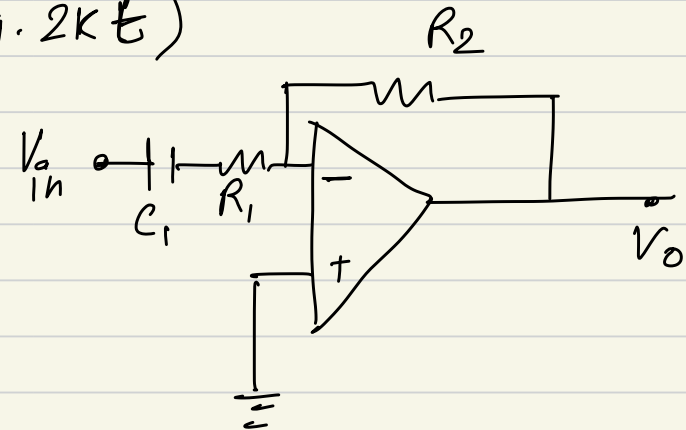
3. (3 points) If you immediately feed the digital output into a 2-bit Inverting R-2R Ladder DAC to generate an analog alarm voltage, where logic levels of Logic 1 = 5 V and Logic 0 = 0 V. The DAC's Op-Amp has a feedback resistor R_f equal to $2R$. Calculate the final analog output voltage (V_{out}) of the DAC.

Tutorial Quiz 9

$$V_{in} = \sin(2\pi \cdot 2t) + 1.5 \sin(2\pi \cdot 2Kt)$$

$$\frac{V_o}{V_{in}} = \frac{-R_2}{R_1 + \frac{1}{j\omega C_1}}$$

$$= \frac{-R_2 j\omega C_1}{1 + j\omega C_1 R_1}$$



$$\Rightarrow \left| \frac{V_o}{V_{in}} \right| = \frac{R_2 C_1 \omega}{\sqrt{1 + (\omega/\omega_c)^2}} \quad \uparrow \text{ where } \omega_c = \frac{1}{R_1 C_1} \quad \text{or } f_c = \frac{1}{2\pi R_1 C_1}$$

for $\omega \ll \omega_c$

$$\left| \frac{V_o}{V_{in}} \right| = R_2 C_1 \omega$$

at $\omega = \omega_c$

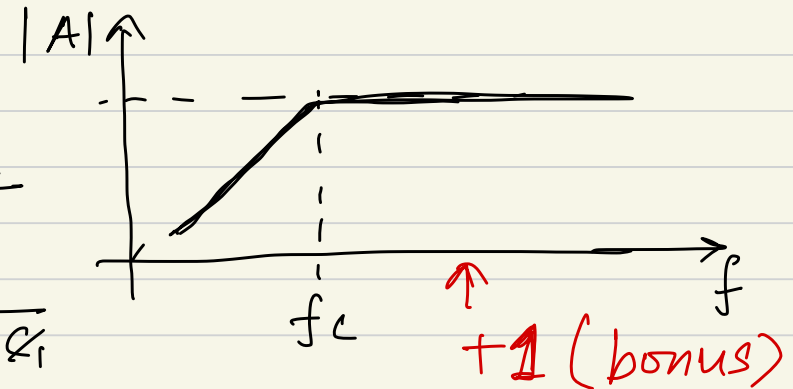
$$\left| \frac{V_o}{V_{in}} \right| = \frac{R_2 C_1 \omega_c}{\sqrt{2}}$$

$$= \frac{R_2 C_1}{\sqrt{2}} \times \frac{1}{R_1 C_1}$$

$$= \frac{R_2}{\sqrt{2} R_1}$$

for $\omega \gg \omega_c$

$$\left| \frac{V_o}{V_{in}} \right| = R_2 C_1 \omega_c = \frac{R_2}{R_1}$$



Given $R_2 = 3.18 \text{ k}\Omega$
 $R_1 = 1.59 \text{ k}\Omega$
 $C_1 = 1 \mu\text{F}$

$\Rightarrow$ maximum gain $|A| = 2 \text{ V/V}$ 1

$$f_c = \frac{1}{2\pi R_1 C_1} = \frac{1}{2\pi \times 1.59 \times 10^3 \times 10^{-6}}$$

$$= \frac{10^3}{2\pi \times 1.59} \approx 100 \text{ Hz}$$
1

therefore, 2Hz signal will be filtered while 2KHz signal will pass with a gain of 2 V/V

$\Rightarrow$ $V_o = -2 \times 1.5 \sin(2\pi \cdot 2000t) = -3 \text{ V} \sin(2\pi \times 2000t)$

1
 inverting
 Amplifier
 (-0.5 if sign is not correct)

$$|V_{o \max}| = 3 \text{ V}$$
1

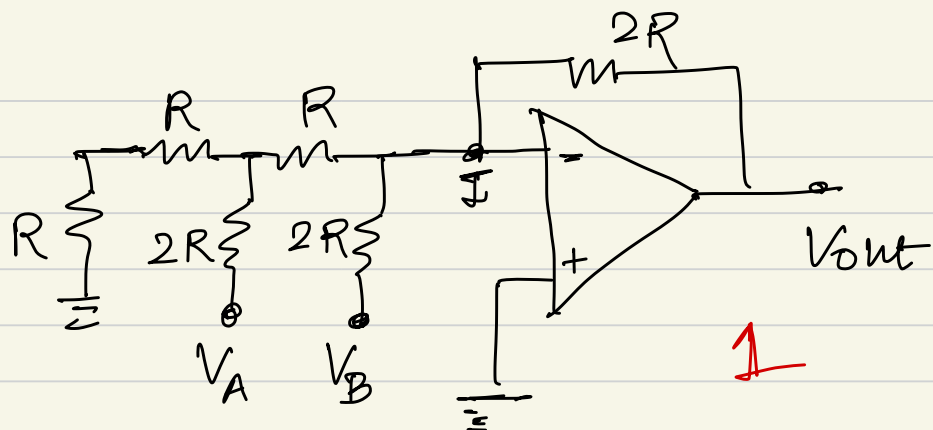
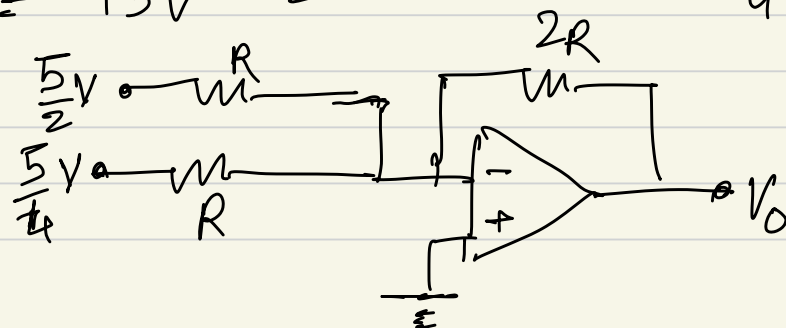
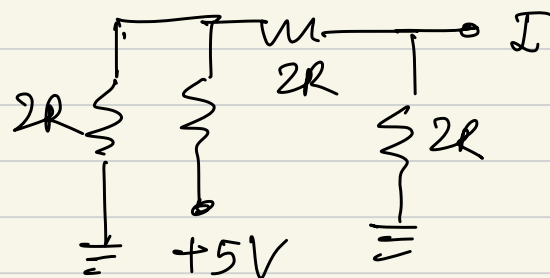
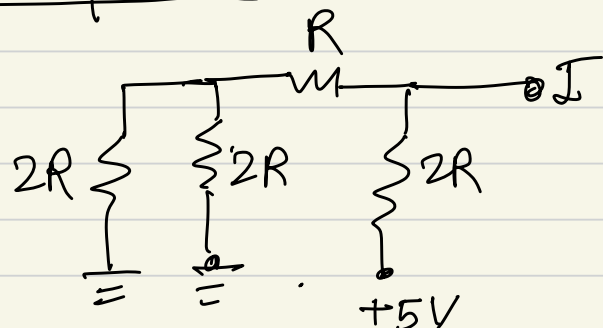
2. A. $LSB = \frac{V_{\text{range}}}{2^N} = \frac{4 \text{ V}}{2^2} = 1 \text{ V}$ 1

B. i.e. $0 \text{ V} \rightarrow 00$, $1 \text{ V} \rightarrow 01$, $2 \text{ V} \rightarrow 10$, $3 \text{ V} \rightarrow 11$ 0.5 3 V $\rightarrow$ 11 Digital Output = 11 0.5

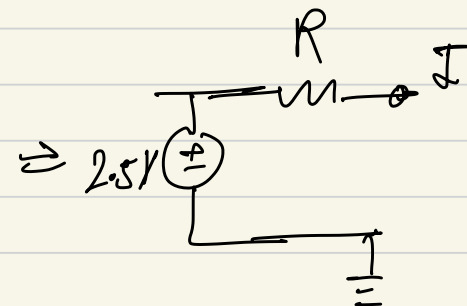
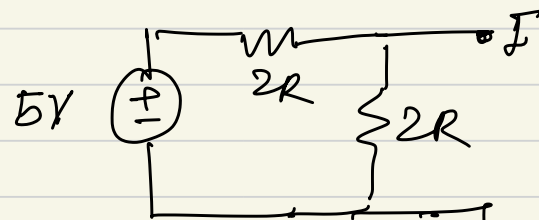
3.

V_A V_B
5V 5V as if 11

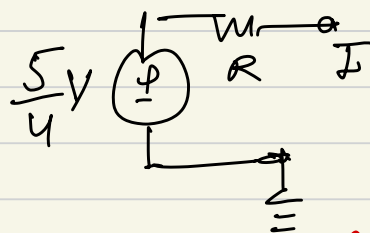
Superposition



if $V_A = 0$, $V_B = +5V$



if $V_A = +5V$, $V_B = 0$



$$V_O = \left(-2 \times \frac{5}{2} - 2 \times \frac{5}{4} \right) = -\frac{10}{2} \times \frac{3}{2} = -7.5V$$